

1.9.3 – 1.9.5 Wireframing, Figma, and Prototypes

وائر فریم یو آئی کا بنیادی خاکہ ہے، رنگوں سے پہلے۔

CS XII

Chapter 1: Computer Systems (HCI)

Lecture 13 of 15

1.9.3 – 1.9.5 Wireframing, Figma, and Prototypes

LEARN IT

1.9.3 Wireframing is creating a basic visual outline or blueprint of a user interface **before** the final design. It plans the structure without focusing on colours or detailed graphics.

Low-fidelity wireframes (Lo-Fi)

A simple black-and-white sketch showing basic layout (boxes and lines). Used early to test and share ideas quickly. Focuses purely on structure and flow.

High-fidelity prototypes (Hi-Fi)

A detailed version that looks almost exactly like the final product. Includes real colours, fonts, images and interactive clickable buttons. Used for final user testing.

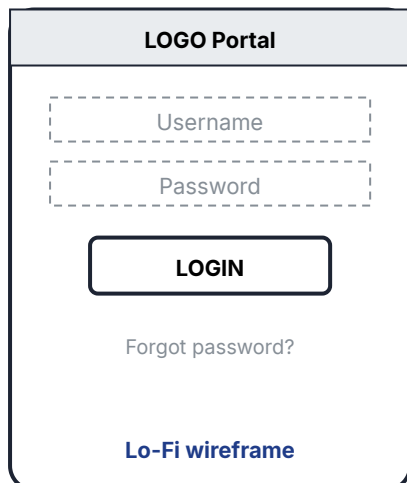
Lo-Fi sketch



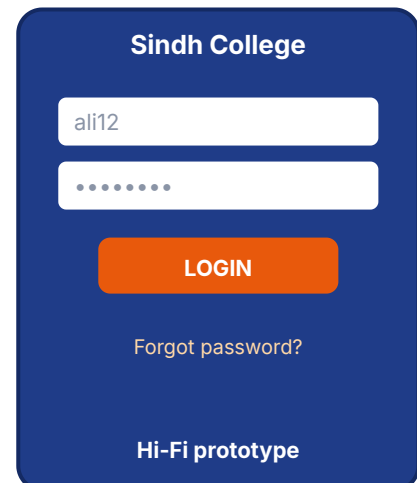
Add details



Hi-Fi prototype



→
add colour



A school-portal login: Lo-Fi boxes and labels on the left; a coloured, realistic Hi-Fi prototype on the right.

1.10 What is Figma?

Figma is a popular, free, web-based tool used to design and collaborate on UI and prototypes. It works directly in a web browser — you do not have to install a heavy program on the college lab PCs.

Creating a low-fidelity design in Figma

1. **Sign up:** access the Figma workspace online.
2. **Create frames:** select the Frame tool (for example an iPhone-sized frame) to act as your screen canvas.
3. **Add shapes:** draw basic rectangles for buttons and circles for avatars.
4. **Add text:** use the text tool to label buttons (for example, "Sign Up").
5. **Duplicate and edit:** copy the frame to create the next screen (for example, the Home Screen).

1.9.5 Prototype design

A **prototype** is an early interactive model used to test design ideas **before writing actual code**. In Figma you use the Prototype tab to link screens together.

- Set a **Trigger** (for example, "On Tap").
- Set an **Action** (for example, "Navigate to Home Screen").

Real users can then click through the fake app to see if the navigation makes sense.

WORKED EXAMPLE

Worked example: JazzCash-style "Send Money" in Figma, without writing code.

1. Lo-Fi on paper or in Figma: three frames — Home (boxes: Send, Bills, Load), Amount (text field + Next), Success (tick + Done).
2. Show the sketch to two classmates. If they cannot find Send, move that box to the top. This is the point of Lo-Fi: cheap changes.
3. In Figma, add a phone frame, rectangles, and labels. Duplicate the frame for Home, Amount and Success.
4. Prototype tab: Trigger = On Tap of the Send rectangle; Action = Navigate to Amount. On Tap of Next → Success.
5. Hand the laptop to a student. If they tap Bills by mistake, the navigation is still wrong — fix it before any programmer writes Java or Kotlin.

REMEMBER IT

Lo-Fi = boxes. Hi-Fi = almost the real app. Prototype = clickable. Figma can hold all three stages. Trigger = what the user does. Action = where the fake app goes.

COMMON MISTAKE

Mistake: Colouring a Lo-Fi wireframe, or thinking a Figma prototype is the finished coded app.

Fix: Lo-Fi stays black-and-white on purpose so people discuss structure, not taste in colours. A prototype tests ideas before code; developers still have to build the real system afterwards.

اردو میں سمجھیں

وائر فریم یو آئی کا بنیادی خاکہ ہے، رنگوں سے پہلے۔ لو فائی سیاہ و سفید ڈبے ہیں؛ ہائی فائی تقریباً حتمی شکل ہے، رنگ، فونٹ اور کلک ہونے والے بٹن کے ساتھ۔ فگما (Figma) مفت ویب ٹول ہے: فریم بناؤ، شکل اور متن ڈالو، فریم کاپی کرو۔ پروٹوٹائپ کوڈ سے پہلے کا چلنے والا نمونہ ہے جس میں ٹرگر (On Tap) اور ایکشن (اگلی اسکرین) جوڑے جاتے ہیں۔

CHECK YOURSELF

1. What does a Lo-Fi wireframe deliberately leave out?
2. Where does Figma run?
3. In Figma prototyping, what is a trigger? Give the source example.

Answers: 1. Colours and detailed graphics (it is structure only). 2. Directly in a web browser.
3. The user event that starts a move between screens, e.g. "On Tap".

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A wireframe is:

- a. A low-detail blueprint of screens
- b. The final branded mock
- c. A running MVP
- d. A database dump

MCQ 2. Figma is:

- a. A collaborative UI/prototype tool in the browser
- b. A DBMS
- c. An OSI sniffer
- d. A sort

MCQ 3. A prototype here is:

- a. A clickable model of the UI
- b. The production server
- c. A primary key
- d. A pie

MCQ 4. You wireframe:

- a. Before polishing pixels
- b. After launch only
- c. Instead of need analysis
- d. Instead of testing ever

MCQ 5. Low fidelity means:

- a. Boxes and labels, not final art
- b. Photoreal 3D
- c. Compiled ARM
- d. Encrypted SQL

MCQ 6. Figma is taught because the book uses it to:

- a. Design and share UI without installing heavy tools
- b. Replace Python
- c. Replace ER
- d. Replace SDLC

Extra short questions

1. Define wireframe and prototype.
2. Why Figma (from the notes)?
3. Order: need analysis → wireframe → prototype → test.
4. What belongs on a wireframe?

Extra long question

1. Describe three Figma wireframe frames for college attendance: login, mark, confirm. List what you would test with five students.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Wireframe: structure/layout. Prototype: interactive simulation of the UI.
2. Free, web-based, collaborative, good for class.
3. Do not skip to colour before structure.
4. Boxes, headings, buttons, notes — not lorem-pretty photos.

Long

1. Frame contents; tap targets; Urdu. Test: time to mark, errors, could they undo. Change the prototype, not the database first.

Next lecture: 1.11 Test and Evaluate HCI (Evaluation Methods)